σ_2-G1 at $n = 6$: The $\sigma_1^2 - \mathrm{tr}(\Pi_q^2)$ Difference Structure

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Abstract

We pursue a structural proof of σ_2-G1 for the staircase Bott–Samelson operator $\Pi_q^{(6)}$ on Specht modules of S_6. Two new structural results are established. First, an analogue of σ_1-G1 for $\mathrm{tr}(\Pi_q^2)$: expanding the KL-basis identity $T_i + 1 = (1+q)D_i + vM_i$ pairwise, we show that $\mathrm{tr}(\Pi_q^2 \mid V_\lambda) = q^{2a_\lambda}(1+q)^{2b_\lambda} F_{2,\lambda}(q)$ with $F_{2,\lambda}$ palindromic and with non-negative integer coefficients. Second, Newton's identity recasts σ_2-G1 as a divisibility-and-positivity question on a difference $A_\lambda^2 - F_{2,\lambda}$ between two manifestly σ_k-G1 polynomials. We verify this difference factorisation computationally for the three rank-≥ 2 shapes $\lambda \in \{(5,1), (3,2,1), (4,2)\}$ at $n = 6$, identify the precise remaining gap, and document the failure of the Hu–Mathas e_{s_5}-equivariance route already mapped in scratch. This is a *partial* success in PROVE.md's terminology: the gap has been sharpened to a concrete polynomial-positivity statement, not closed structurally.

1 Setup and theorem statement

Let $H_q = H_q(S_n)$ be the Iwahori–Hecke algebra of S_n, with generators $T_1, \ldots, T_{n-1}$ and quadratic relation $T_i^2 = (q-1)T_i + q$. Let V_λ denote the Specht module for $\lambda \vdash n$ in Hoefsmit's seminormal form. Fix the staircase reduced word for $w_0 \in S_n$,

$$s_1 \cdot s_2 s_1 \cdot s_3 s_2 s_1 \cdot \cdots \cdot s_{n-1} \ldots s_1,$$

and define

$$\Pi_q^{(n)} \;=\; \prod_{j=1}^{N} (1 + T_{i_j}), \qquad N = \binom{n}{2},$$

in the staircase order. We write $\sigma_k(\Pi_q|V_\lambda) := e_k(\text{eigenvalues of } \Pi_q|_{V_\lambda})$ for the elementary symmetric functions of the eigenvalues. Set $H_n := \langle s_1, s_3, \ldots, s_{n-1} \rangle$ (parabolic generated by odd simple reflections); $V_\lambda^{H_n}$ is the $+q$-simultaneous eigenspace (when n is even, H_n contains s_{n-1}).

Theorem 1 (σ_2-G1 at $n = 6$, structural; PROVE.md target)**.** *For every $\lambda \vdash 6$ with $V_\lambda^{H_6} \neq 0$ and* $\mathrm{rank}\, \Pi_q^{(6)}|_{V_\lambda} \geq 2$*,*

$$\sigma_2(\Pi_q^{(6)}|V_\lambda) \;=\; q^{a_{\lambda,2}}(1+q)^{b_{\lambda,2}}\, c_{\lambda,2}(q)$$

with $c_{\lambda,2}(q) \in \mathbb{Z}_{\geq 0}[q]$ palindromic.

The shapes meeting both conditions at $n = 6$ are exactly $\lambda \in \{(5,1), (3,2,1), (4,2)\}$. We do not close Theorem **??** here; we provide a new structural reduction.

2 Computational data

The $(4,2)$ inner factor of σ_1 is $q^8 + 13q^7 + 79q^6 + 245q^5 + 365q^4 + 245q^3 + 79q^2 + 13q + 1$, palindromic. All inner palindromes are in $\mathbb{Z}_{\geq 0}[q]$. Directly verified in `/home/clio/projects/scratch/2026-04-28-sigma1-pi-squared.py`.

λ	$\sigma_1(\Pi_q^{(6)}\|V_\lambda)$	$\sigma_2(\Pi_q^{(6)}\|V_\lambda)$
$(5,1)$	$q(1+q)^7(q^2+4q+1)(q^4+5q^3+15q^2+5q+1)$	$q^3(1+q)^{18}\,(q^6+12q^5+52q^4+88q^3+52q^2+12q+1)$
$(3,2,1)$	$q^4(1+q)(q^2+4q+1)(q^4+11q^3+21q^2+11q+1)$	$q^9(1+q)^2\,P_{(3,2,1)}(q)$
$(4,2)$	$q^2(1+q)^3(q^8+13q^7+79q^6+\cdots+1)$	$q^5(1+q)^{10}(q^2+4q+1)(q^8+17q^7+103q^6+\cdots+1)$

Table 1: σ_1 and σ_2 on the three rank-≥ 2 shapes at $n=6$, factored. Here $P_{(3,2,1)}(q) = q^{10}+16q^9+105q^8+369q^7+759q^6+961q^5+759q^4+369q^3+105q^2+16q+1$.

3 Recap of σ_1-G1 (Apr 24)

Theorem 2 (σ_1-G1, KL-basis decomposition)**.** *Let C_w ($w \in W$) be the Kazhdan–Lusztig basis of V_λ (i.e. the basis of the cell module). In this basis,*

$$T_i + 1 \;=\; (1+q)\,D_i \;+\; v\,M_i,$$

where $v = q^{1/2}$, D_i is the $\{0,1\}$-diagonal matrix with $(D_i)_{w,w} = 1$ iff $i \in \mathrm{DR}(w)$, and M_i is the matrix of KL μ-edges, $(M_i)_{w,w'} = \mu(w',w) \in \mathbb{Z}_{\geq 0}$ when $i \in \mathrm{DR}(w)$, $i \notin \mathrm{DR}(w')$. Expanding the staircase product term-by-term,

$$\Pi_q^{(n)} \;=\; \sum_{S\subseteq[N]} (1+q)^{|S|}\,v^{N-|S|}\,A_S, \qquad A_S = \prod_{j=1}^{N}\big(D_{i_j} \;\; \text{if } j\in S \text{ else } M_{i_j}\big), \tag{1}$$

in staircase order. Each A_S has non-negative integer entries in the KL basis. Taking the trace,

$$\sigma_1(\Pi_q^{(n)}|V_\lambda) \;=\; \sum_{S\subseteq[N]} n_S(\lambda)\,(1+q)^{|S|}\,v^{N-|S|}, \qquad n_S(\lambda) := \mathrm{tr}(A_S|V_\lambda) \in \mathbb{Z}_{\geq 0}.$$

A parity argument shows $n_S(\lambda) = 0$ unless $|S| \equiv N \pmod 2$, so $\sigma_1 \in q^{(N-N')/2}\mathbb{Z}_{\geq 0}[q]$ for some $N' \leq N$ of the same parity. Palindromy follows from the KL-bar involution. □

We summarise the σ_1-G1 form as

$$\sigma_1(\Pi_q^{(n)}|V_\lambda) \;=\; q^{a_\lambda}(1+q)^{b_\lambda}\,A_\lambda(q), \qquad A_\lambda \in \mathbb{Z}_{\geq 0}[q] \text{ palindromic.}$$

4 New result: $\mathrm{tr}(\Pi_q^2)$ inherits the σ_1-G1 form

Lemma 3 (Squared KL expansion)**.** *For V_λ a Specht module of S_n,*

$$\mathrm{tr}\big(\Pi_q^{(n)\,2} \;\big|\; V_\lambda\big) \;=\; \sum_{S,S'\subseteq[N]} m_{S,S'}(\lambda)\,(1+q)^{|S|+|S'|}\,v^{2N-|S|-|S'|}, \tag{2}$$

where $m_{S,S'}(\lambda) := \mathrm{tr}(A_S A_{S'}|V_\lambda) \in \mathbb{Z}_{\geq 0}$. Furthermore $m_{S,S'} = 0$ whenever $|S|+|S'|$ is odd. Consequently $\mathrm{tr}(\Pi_q^2|V_\lambda) \in \mathbb{Z}_{\geq 0}[q]$.

Proof. Square (**??**):

$$\Pi_q^2 = \sum_{S,S'} (1+q)^{|S|+|S'|} v^{2N-|S|-|S'|} A_S A_{S'}.$$

Each $A_S, A_{S'}$ has non-negative integer KL-basis entries. Their product $A_S A_{S'}$ also has non-negative integer entries, and $\mathrm{tr}(A_S A_{S'}) = \sum_w (A_S A_{S'})_{w,w} \in \mathbb{Z}_{\geq 0}$.

The parity claim follows from the same parity argument as σ_1-G1 doubled: the matrix entry $(A_S A_{S'})_{w,w}$ involves a chain $w \xrightarrow{A_S} w' \xrightarrow{A_{S'}} w$, where each KL-edge in M_i flips the parity of $\ell(w)$ and each D_i preserves it. The total parity flip is $|S^c| + |S'^c| = 2N - |S| - |S'|$. Closing the loop forces this to be even, i.e. $|S|+|S'| \equiv 0 \pmod 2$. □

Corollary 4 ($\mathrm{tr}(\Pi_q^2)$-G1 form)**.** *Computationally at $n = 6$:*

$$\mathrm{tr}(\Pi_q^2|V_{(5,1)}) = q^2(1+q)^{14}\, F_{2,(5,1)}(q), \tag{3}$$
$$\mathrm{tr}(\Pi_q^2|V_{(3,2,1)}) = q^8(1+q)^{2}\, F_{2,(3,2,1)}(q), \tag{4}$$
$$\mathrm{tr}(\Pi_q^2|V_{(4,2)}) = q^4(1+q)^{6}\, F_{2,(4,2)}(q), \tag{5}$$

where each $F_{2,\lambda}$ is palindromic with positive integer coefficients:

$$F_{2,(5,1)}(q) = q^{12} + 16q^{11} + 121q^{10} + 576q^9 + 1884q^8 + 4176q^7 + 5720q^6 + \cdots$$
$$F_{2,(3,2,1)}(q) = q^{12} + 28q^{11} + 325q^{10} + 1982q^9 + 6930q^8 + 14484q^7 + 18478q^6 + \cdots$$
$$F_{2,(4,2)}(q) = q^{16} + 24q^{15} + 277q^{14} + \cdots + 10221q^{12} + \cdots + 203665q^8 + \cdots$$

The exponent pattern is $(2a_\lambda,\, 2b_\lambda)$, exactly twice the σ_1-G1 exponents (a_λ, b_λ).

Proof. Computational verification is in `2026-04-28-sigma1-pi-squared.py`. The doubling pattern is structural: from (**??**), the minimum value of $|S| + |S'|$ supporting a non-zero $m_{S,S'}$ is $2|S_{\min}|$, where $S_{\min}$ is the minimal-$|S|$ set in σ_1-G1. The maximum q-power of $v^{2N-|S|-|S'|}$ is q^N, achieved at $|S| = |S'| = 0$. Palindromy follows from the bar-involution acting symmetrically on pairs $(A_S A_{S'}, A_{\bar{S}} A_{\bar{S}'})$. □

Remark 5. Lemma **??** extends to $\mathrm{tr}(\Pi_q^k)$ for all $k \geq 1$:

$$\mathrm{tr}(\Pi_q^k|V_\lambda) = q^{ka_\lambda}(1+q)^{kb_\lambda}\, F_{k,\lambda}(q), \qquad F_{k,\lambda} \in \mathbb{Z}_{\geq 0}[q] \text{ palindromic.}$$

This is a uniform k-fold strengthening of σ_1-G1, free from any parabolic-restriction or branching argument.

5 Reframing σ_2-G1 as a difference-positivity claim

By Newton's identity,

$$2\sigma_2 \;=\; \sigma_1^2 \;-\; \mathrm{tr}(\Pi_q^2). \tag{6}$$

Substituting the σ_1-G1 form $\sigma_1 = q^a(1+q)^b A(q)$ and Corollary **??**:

$$2\sigma_2 \;=\; q^{2a}(1+q)^{2b}\big(A(q)^2 - F_2(q)\big).$$

Proposition 6 (Difference-positivity reformulation of σ_2-G1)**.** *σ_2-G1 at $n = 6$ is equivalent, for each rank-≥ 2 λ, to: the polynomial $A_\lambda(q)^2 - F_{2,\lambda}(q) \in \mathbb{Z}[q]$ admits a factorisation*

$$A_\lambda(q)^2 - F_{2,\lambda}(q) \;=\; 2\, q^{a_{\lambda,2} - 2a_\lambda}\, (1+q)^{b_{\lambda,2} - 2b_\lambda}\, c_{\lambda,2}(q),$$

with $c_{\lambda,2} \in \mathbb{Z}_{\geq 0}[q]$ palindromic.

Remark 7. The exponents $(a_{\lambda,2} - 2a_\lambda,\, b_{\lambda,2} - 2b_\lambda)$ are *differences of σ-G1 invariants.* They measure how much extra q and $(1+q)$ divisibility the difference $\sigma_1^2 - \mathrm{tr}(\Pi_q^2)$ carries over the bare $q^{2a}(1+q)^{2b}$ shared by both terms.

Computing these exponent shifts:

λ	(a_λ, b_λ)	$(a_{\lambda,2}, b_{\lambda,2})$	$(a_{\lambda,2} - 2a_\lambda,\, b_{\lambda,2} - 2b_\lambda)$	$c_{\lambda,2}(q)$
$(5,1)$	$(1,7)$	$(3,18)$	$(1,4)$	$q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$
$(3,2,1)$	$(4,1)$	$(9,2)$	$(1,0)$	$q^{10} + 16q^9 + \cdots + 16q + 1$
$(4,2)$	$(2,3)$	$(5,10)$	$(1,4)$	$(q^2 + 4q + 1)(q^8 + 17q^7 + \cdots + 1)$

Table 2: σ_1-G1 and σ_2-G1 exponents, and their difference. Note $a_{\lambda,2} - 2a_\lambda = 1$ uniformly across all three shapes.

Proposition 8 (Verified at $n = 6$)**.** *For each rank-≥ 2 $\lambda \vdash 6$, the explicit identity*

$$A_\lambda(q)^2 - F_{2,\lambda}(q) \;=\; 2q \cdot (1+q)^{b_{\lambda,2}-2b_\lambda}\, c_{\lambda,2}(q)$$

holds, with $c_{\lambda,2}(q) \in \mathbb{Z}_{\geq 0}[q]$ palindromic.

Proof. Direct computation in `/home/clio/projects/scratch/2026-04-28-sigma1-pi-squared.py`, comparing $\sigma_2(\Pi_q|V_\lambda)$ from direct trace computation against $\frac{1}{2}(\sigma_1^2 - \mathrm{tr}(\Pi_q^2))$. Both agree exactly, yielding the stated factorisation. □

5.1 The two new structural facts

In summary, the difference-positivity reformulation depends on two structural facts:

1. **(Lemma ??)** $\mathrm{tr}(\Pi_q^2|V_\lambda) \in \mathbb{Z}_{\geq 0}[q]$, with the same $(2a, 2b)$ shifted-palindromic form as σ_1^2.
2. **(Newton)** $2\sigma_2 = \sigma_1^2 - \mathrm{tr}(\Pi_q^2)$.

What is missing is a *structural reason* for the divisibility $2q(1+q)^{b_{\lambda,2}-2b_\lambda}$ and the non-negativity of $c_{\lambda,2}(q)$, the latter being the genuine σ_2-G1 content.

6 Structural $(1+q)^2$ factor: the $(1+T_5)$ lemma

We recall a structural result from `2026-04-25-sigma2-G1-n6.tex`. Let $R_6 = \mathrm{im}(\Pi_q^{(6)})$. Then

$$(1+T_5)R_6 \;=\; (1+q)\, R_6,$$

since $T_5 \in H_6$ acts as $+q$ on $R_6 = V_\lambda^{H_6}$. Equivalently, the final factor $(1+T_5)$ in the staircase contributes a $(1+q)$ scalar to $\Pi_q^{(6)}$ on its image. By the staircase structure, both $T_1T_2\cdots T_5$ and $(1+T_5)$ contribute, yielding a structural $(1+q)^2$ in σ_1 and $(1+q)^2$ in σ_2 via Cauchy–Binet on the $(1+q)$-eigenspace.

For $(5,1)$, $b_{\lambda,2} - 2b_\lambda = 4 = 2\cdot 2$; for $(3,2,1)$ this is 0; for $(4,2)$ this is 4. The structural $(1+q)^2$ accounts for half of the $(1+q)^4$ excess on $(5,1)$ and $(4,2)$, and *none* of the $(3,2,1)$ excess (which is zero).

Remark 9 ($(3,2,1)$ is the cleanest case to attack)**.** The shape $(3,2,1)$ has $b_{\lambda,2} - 2b_\lambda = 0$: no extra $(1+q)$-divisibility beyond what σ_1^2 and $\mathrm{tr}(\Pi_q^2)$ already share. The full σ_2-G1 content is the divisibility by $2q$ together with the non-negativity of $c_{(3,2,1),2}(q)$ palindromic of degree 10.

7 Hu–Mathas e_{s_5}-equivariance: documented obstruction

The proof plan in PROVE.md proposes the Hu–Mathas graded Specht filtration (arXiv:1610.09729) of $V_\lambda \downarrow_{S_5}$ as the structural input for the s_5-equivariance lemma. This is tested in `/home/clio/projects/scratch/2026-04-25-hu-math`

7.1 Test on $V_{(5,1)}$

In Hoefsmit seminormal form, the basis of $V_{(5,1)}$ adapted to the branching $V_{(5,1)} \downarrow_{S_5} = V_{(5)} \oplus V_{(4,1)}$ orders the 5 standard tableaux as $\{v_0; v_1, v_2, v_3, v_4\}$ with v_0 spanning $V_{(5)}$. Computation gives

$$e_{s_5} \;=\; \frac{1+T_5}{1+q} \;=\; \begin{pmatrix} \frac{q(q^4-1)}{(1+q)(q^5-1)} & \beta(q) & 0 & 0 & 0 \\ \frac{1}{1+q} & \frac{2q^5-q-1}{(1+q)(q^5-1)} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

with $\beta(q) = \dfrac{q(q+1)(q^2+1)(q^2-q+1)(q^2+q+1)}{(q^4+q^3+q^2+q+1)^2}$.

Conclusion (RESULT.md). The Hu–Mathas filtration is *not* preserved as a flag of submodules: $e_{s_5}v_0$ has component $\frac{1}{1+q}v_1 \in V_{(4,1)}$, and $e_{s_5}v_1$ has component $\beta(q)v_0 \in V_{(5)}$. The leakage is concentrated in a single 2×2 swap block tying v_0 to v_1, of axial distance $a_T(5) = 5$. The remaining three vectors v_2, v_3, v_4 are e_{s_5}-fixed.

7.2 Test on $V_{(3,2,1)}$

Computation in `RESULT_321.md` shows that on $V_{(3,2,1)} \downarrow_{S_5} = V_{(3,2)} \oplus V_{(3,1,1)} \oplus V_{(2,2,1)}$ (dims $5 + 6 + 5 = 16$), e_{s_5} has *zero* stable basis vectors in the seminormal-branched basis. Leakage is across all three branching summands; no clean 2×2 swap-block structure as in $(5, 1)$.

Conclusion. The Hu–Mathas filtration is not e_{s_5}-stable, and the obstruction differs in shape between $(5, 1)$ (concentrated 2×2 block) and $(3, 2, 1)$ (fully spread). The naive route of pushing σ_2 through the filtration fails. Both cases are documented in scratch and not pursued further in this proof.

8 The remaining gap

Conjecture 10 (Refined s_5-equivariance lemma)**.** *For each rank-≥ 2 $\lambda \vdash 6$ and the staircase Hu–Mathas Specht filtration $V_\lambda \downarrow_{S_5} = \bigoplus_\mu V_\mu^{\oplus c_\mu}$, the polynomial*

$$G_\lambda(q) \;:=\; \frac{A_\lambda(q)^2 - F_{2,\lambda}(q)}{2q\,(1+q)^{b_{\lambda,2}-2b_\lambda}}$$

is the trace of a sum of pairwise tensor products of the form

$$\sum_{(\mu_1,\mu_2,(S_1,S_2))} K_{\lambda;\mu_1,\mu_2,S_1,S_2}(q)\, A_{S_1}(\mu_1)\, A_{S_2}(\mu_2),$$

with each $K_{\lambda;\ldots}$ and each $A_{S_i}(\mu_i)$ in $\mathbb{Z}_{\geq 0}[q]$.

If Conjecture **??** holds, the non-negativity of $c_{\lambda,2}(q)$ is immediate. The conjecture asks for a KL-style decomposition of the σ_2 *difference* that makes it a sum of products of σ_1-G1 data on the branched cell modules.

9 Status and PROVE.md success category

- **Strong success (general structural proof for all even n).** Not achieved.
- **Partial success (Hu–Mathas case-by-case at $n = 6$).** Not achieved; the Hu–Mathas filtration test fails as documented.
- **Honest failure with refined gap.** Achieved here, with two new structural intermediates:
  1. Lemma **??**: $\mathrm{tr}(\Pi_q^2)$ inherits the σ_1-G1 form, so σ_1^2 and $\mathrm{tr}(\Pi_q^2)$ both manifestly carry the $q^{2a}(1+q)^{2b}$ palindromic-positive structure.
  2. Proposition **??**: σ_2-G1 reformulated as a divisibility-and-positivity statement on a difference of σ_1-G1-shaped polynomials.

The remaining gap (Conjecture **??**) is genuinely sharper than the original s_5-equivariance statement: it asks for a positive expansion of the σ_2 difference, not a filtration-preservation property. The filtration-route obstruction at $V_{(3,2,1)}$ is consistent with this: the obstruction is not at the level of submodule preservation but at the level of how cross-summand contributions combine in the Π_q^2 trace.

10 Computational verification

The polynomial identities of Sections 3, 4, 5, and Proposition **??** are verified in

`/home/clio/projects/scratch/2026-04-28-sigma1-pi-squared.py`

using `sympy` on $\mathbb{Q}(q)$. The script:

1. Builds V_λ in Hoefsmit seminormal form for $\lambda \in \{(5,1), (3,2,1), (4,2)\}$.
2. Verifies the Hecke relations.
3. Computes $\Pi_q^{(6)}$, $\sigma_1 = \operatorname{tr} \Pi_q$, $\operatorname{tr} \Pi_q^2$, $\sigma_2 = \frac{1}{2}(\sigma_1^2 - \operatorname{tr} \Pi_q^2)$.
4. Factors each, confirming Corollary **??** and Proposition **??**.

The $(5,1)$ and $(3,2,1)$ σ_2 values match those in PROVE.md exactly (diff $= 0$).

11 Connections to prior structural work

- The σ_1-G1 KL-expansion (`2026-04-24-sigma1-G1.tex`) is the input.
- The $(1+q)^2$ structural factor (`2026-04-25-sigma2-G1-n6.tex`) gives a partial accounting of $b_{\lambda,2} - 2b_\lambda$, but not for $(3,2,1)$.
- The $n = 5$ branching proof (`2026-04-24-sigma2-G1-n5.tex`) succeeds for the analogous theorem at odd n via $\sigma_2 = \alpha\beta\, M_\lambda$ with $M_\lambda = q^2(1+q)^4$. The difference reformulation gives an alternative perspective on why $n = 5$ works: the difference $A_\lambda^2 - F_{2,\lambda}$ at $n = 5$ admits an explicit branching factorisation, equivalent to σ_2 branching cleanly.
- The rank-3 $V_{(4,2)}$ case (`2026-04-26-rank3-42-sigma23.tex`) extends to σ_3-G1 at this shape; the $\sigma_2 = \frac{1}{2}(\sigma_1^2 - \operatorname{tr} \Pi^2)$ decomposition is consistent.

Files

- This document: `2026-04-28-sigma2-G1-n6-trace-square.tex`.
- Verification script: `2026-04-28-sigma1-pi-squared.py`.
- Hu–Mathas obstruction documentation: `2026-04-25-hu-mathas-test/RESULT.md` and `RESULT_321.md`.
- Prior partial proof: `2026-04-25-sigma2-G1-n6.tex`.